Why Domain Matters, part 3

Payments

The record says

`Payment failed.`

What actually happened

The customer’s account was debited.

Saturday 20 June, 6:48 pm. Store 014, Jayanagar.

The till said the payment failed. Her bank had already taken the money.

```
18:48:25  UPI   ₹1,064.47  failed: no response
18:48:55  card  ₹1,064.47  approved
```

87 seconds after she approved the UPI payment, the store was credited. Nobody at the till saw it. She had paid twice.

Five readings of one payment

Data product	What it reads	What it concludes
Payments dashboard	Approvals at the tills 95.9% for June; 69% and 67% in the 18:00 and 19:00 hours on 20 June, normal by eight	“Incident closed.”
Data tests	10 of 10 pass: one approved payment per bill, every approval a success at the provider, settlements add up	“The pipeline is green.”
Reconciliation report	607 credits against 597 approvals on Saturday, 670 against 663 on Sunday	“Park the 17 in suspense.”
Customer support	26 payment complaints in June, 23 of them “charged twice”; ₹11,302 of refunds	“Screenshot, check, refund.”
AI assistant, given the till’s schema	All 30,077 bills have exactly one approved payment	“Nobody paid twice.”

Each conclusion follows from the records in front of it. None of them sees a customer charged twice.

A slow answer is not a failed payment

Her app sends the payment → **NPCI** routes it → **Her bank** debits her account → **Store's bank** slow to answer → **The till** gives up at 30 s

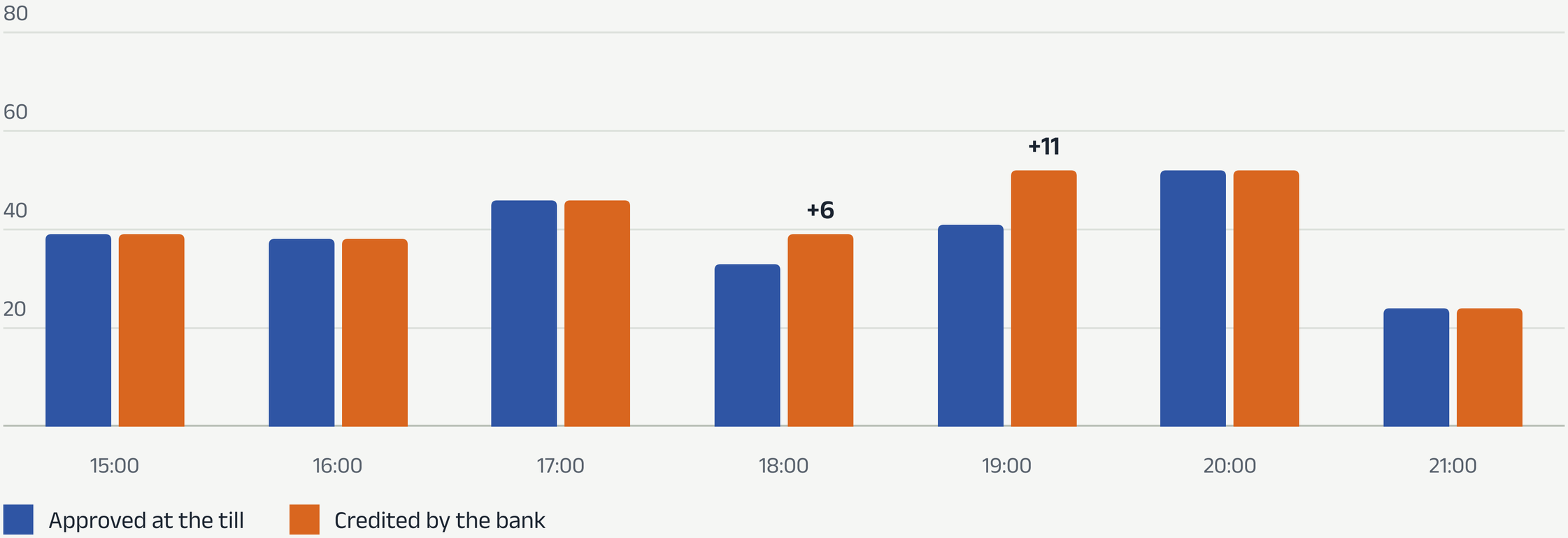
When the debit has happened and the answer is late, UPI settles it afterwards: the store's bank confirms the credit, or the money goes back. RBI requires it back within five days of the payment, with ₹100 a day after that. The money is never lost. For a while it is in two places, and the till is the last to know.

28 of 82 UPI attempts at Store 014 got no answer within 30 seconds between 18:20 and 19:50. RBI/2019-20/67 (2019): merchant payments reversed within T+5.

Every record was right about what it saw

The till	`18:48:25 UPI ₹1,064.47 failed: no response at 18:48:55`
The till	`18:48:55 card ₹1,064.47 approved`
Provider’s report	`18:49:52 UPI 617118013344 success`
Settlement	`Sun 21 Jun: credited with Saturday’s UPI payments`
Customer support	`Tue 23 Jun, 14:00: “charged twice”`
Refund	`Sun 28 Jun, 14:00: ₹1,064.47 back`
Her money	Taken twice, for eight days

The bank was paid for more than the tills approved



UPI payments at Store 014's tills, Saturday 20 June 2026, by hour. +N: credited by the bank, marked failed at the till. Synthetic world, seed 42.

What 90 minutes cost the customers

15

customers charged twice, two of them three times: 17 extra charges, ₹7,826

4.9

median days to a refund, which came only after a complaint

All of June

Customers charged twice	24
Extra charges	26 (₹13,366)
Debited, then returned by their bank	6, within 3 days
Payment complaints	26
Still in suspense when June closed	4 (₹2,846)

Between 18:20 and 19:50 Store 014 took 119 bills; the tills gave up on 28 of 82 UPI attempts.

Nothing in the records says “charged twice”

```
Till, bill B260620-110056
18:48:25  UPI   failed: no response
          reference: none
18:48:55  card  approved

Provider’s UPI report
617118013344  18:48:25
success, confirmed 18:49:52
```

A till that hears nothing back never learns the payment’s reference number, the RRN, so there is nothing to join on.

Data tests check that records agree, and these agree. The only record of the double charge was in the customer’s bank statement.

Identifiers in the data are synthetic. Download June’s till payments and the provider’s report to try the match.

The pipeline is green

10 of 10

data tests pass

Keys, statuses and reconciliations across the till, the provider's report, settlements, refunds and complaints. They agree, because each records only what it saw.

0 of 5

tests about how UPI works pass

- Every success in the provider's report has an approved till payment (fails on 26 credits)
- Every UPI attempt joins the provider's status by RRN (fails on 48 timeouts)
- No bill is paid twice, once timeouts are matched (fails on 24 bills)
- Every extra charge comes back within five days of the debit (fails on 14 charges)
- No hour has a burst of timeouts (fails on 2 hours)

All 15 tests use only the records. The five days are what RBI allows for a failed payment to a merchant, used as the store's own clock.

Treat a timeout as a question

Match each of June's 48 timeouts to the provider row no till payment claims, with the same amount and a start within five seconds. All 48 match:

26

credited to the store late

an extra charge whenever the bill was paid again

6

debited, then returned

by the customer's bank, within days

16

never left the account

the only true failures

> **Then the double charges are a query: late credits on bills paid again**
>
> 26 extra charges on 24 customers, ₹13,366.06: the answer key's 26, with no false alarms. Run each morning on the provider's file, it finds them a median 0.6 days after the debit, against 2 for complaints, and three customers never complained.
>
> `18:48:25 timeout ₹1,064.47 → 617118013344 success 18:49:52 → bill B260620-110056 paid again by card`

Two KPIs, read the way the money moves

Refund wait

3 days

median, from the complaint, as support reports it

4.9 days

median, from the extra debit, when the customer lost the money

10 of 22 refunds took longer than RBI’s five days; four, ₹2,846.06, were not refunded by month end.

UPI decline rate

0.76%

technical declines in June, in line with UPI as a whole; timeouts not counted

25%, 26%

no answer, 18:00 and 19:00 on 20 June, timeouts counted

A monthly rate hides the two hours in which 15 customers paid twice.

Same records, same month. Grey: what the store reports. Red: the clock started at the debit, and the attempts counted the way UPI works.

Give the assistant a domain layer

1 → 7 of 7

questions answered right at the month-end review, schema only → with the domain layer

- “A timeout is a question, not an answer”
- “A provider success with no approved till payment is a customer’s money”
- “Read payment rates by the hour, with timeouts counted”
- The provider’s UPI report, and two views: the timeouts matched, the extra charges

Did any customer pay twice in June? Schema only:

No. All 30,077 bills have exactly one approved payment.

With the domain layer:

Yes: 24 customers, 26 extra charges, ₹13,366.06. Each is a UPI payment the till gave up on and the store’s bank credited later, on a bill the customer then paid again.

The assistant is scripted so the run can be repeated; the technical companion shows every query.

This is the normal case

24.07 bn

UPI transactions in September 2026, about 802 million a day

0.7–0.8%

technical declines, NPCI’s chief executive said in November 2024

90 s

before the first status check NPCI allows on a payment, since its 2025 rules

On 12 April 2025 UPI’s success rate fell to about 50% for nearly two hours, the third disruption in a week. NPCI blamed banks flooding it with status checks, and rationed them.

Sources: NPCI via Outlook Money (2 Oct 2026); Inc42 (20 Nov 2024); MediaNama on NPCI OC 215 (Apr 2025); Outlook Money (16 Apr 2025).

One name, four numbers

61%

The till

UPI attempts the till marked approved. June: 95.9%

82%

The bank

UPI attempts that ended as a credit, late ones included. June: 96.1%

100%

Bills paid

Bills the store was paid for, one way or another. June: 100.0%

87%

Charged once

Bills where no customer’s money moved twice. June: 99.9%

Payment success during the slowdown, Saturday 18:20–19:50. Over the month the four sit within a few points; in the 90 minutes that mattered they spread from 61% to 100%.

Same store, different cost: the new UPI fee

₹8,082

a month for Store 014 at June’s mix: 0.4% on 668 UPI bills above ₹2,000

From 15 October 2026: 0.4% on payments to a merchant above ₹2,000, capped at ₹300. Merchants receiving up to ₹1 lakh a month through UPI QR are exempt.

The fee on one payment

Payment	Fee
₹1,999	₹0.00
₹2,000	₹0.00
₹2,001	₹8.00
₹5,000	₹20.00
₹75,000	₹300.00

NPCI UPI and Services Steering Committee, reported by Deccan Chronicle (15 Sep 2026) and Business Standard (16 Sep 2026).

One change prevents most double charges. The other finds them all.

Wait for the status check

24 → 9

customers charged twice in June

Till time in the slowdown: 77 → 99 minutes, at the worst moment.

Bill ID on every UPI request

4.9 → 0.7

median days to a refund

Nothing left in suspense; complaints 26 → 12. Double charges still happen; every one is found the next morning.

Same customers, baskets and banks in every scenario; the changes start on 1 June. Try them in the demo.

Five questions to ask your team

1. Which of our “failed” statuses are answers, and which are only timeouts?
2. When a key is missing, which other fields can join the records, and who checks the match?
3. Do we measure customer outcomes, such as paying twice, or only system outcomes, such as approvals?
4. Do our clocks start when the customer is harmed or when the customer complains?
5. What does our AI assistant know about how payments settle, and how do we test that?

For data and AI leaders. The technical companion answers each one for Saturday 20 June.

Every record was right about what it saw. Only the customer saw both debits.

The tests in SQL, the matcher and a notebook on the payment records:

kvinayakpai.github.io/domain-series/payments/technical

> **Next: all three on one map**
>
> Who saw what: which records copy each other, and which look at the world.